NYPD Shooting Incident Data (Historic)

Report

2026-10-13

Introduction

Shootings occur in every major municipality in the world, which makes the City of New York no exception. Across the five boroughs and their numerous precincts, NYPD officers react to and produce meticulous notes on each case. This brief report will, therefore, present the main findings from the historical NYPD Shooting Incident Data set, with the newest version published by the City of New York on NYC OpenData [1], which includes every shooting incident recorded by the NYPD since 2006. The text answers three research questions: (1) what are the long-term trends of shooting incidents for the City as a whole and for each borough (with observations aggregated to monthly frequency to employ time-series models), (2) are shootings distributed uniformly across boroughs and precincts, and (3) which geographical or victim characteristics contribute to a fatal outcome (i.e., the incident being declared a murder). Each research question is complemented with a formal statistical hypothesis (or a statistical model) tested at the standard 5% significance level ($\alpha = 0.05$):

- **H1**. Monthly reported incidents have a statistically significant linear trend, i.e., $\beta_1 \neq 0$ in $y_t = \beta_0 + \beta_1 t$.
- **H2**. Within each borough, incidents are distributed uniformly under a Pearson's χ^2 goodness-of-fit test.
- **H3**. The possibility of an incident being classified as a murder depends on precinct, victim age group, sex, and/or race, based on a logistic regression model $\log(\frac{p_i}{1-p_i}) = \beta_0 + \sum_i \beta_i x_i$, in which $\exists_{\beta_i}\ \beta_i \neq 0$.

To compile this report, R packages `tidyverse` and `lubridate` were employed for data processing, alongside `ggplot2` for visualization and the `chisq.test()`, `lm()` and `glm()` functions for statistical modeling. All R code is embedded into the R Markdown document to ensure its reproducibility. The original CSV dataset, downloaded from Coursera, is accessed at `./DATA/NYPD_Shooting_Incident_Data__Historic_.csv`. The report is subdivided into five parts: data description and processing, time trend analysis, spatial structure analysis, (logistic) incident classification model, and discussion of potential data- and method-related bias.

Data Description and Processing

The raw (unprocessed) NYPD Shooting Incident dataset consists of 29,744 rows and 21 columns, corresponding to 23,298 unique incidents (one row per victim) recorded between January 01, 2006 at 02:00 and December 31, 2024 at 19:16. The data cover 5 boroughs (BROOKLYN, BRONX, QUEENS, MANHATTAN, STATEN ISLAND). Only about 19.4% of the records were classified as murders with predominantly male victims (90.2%). Given that the data comprises 94,426 missing values, excessively detailed temporal information (dates and times) for the intended monthly time-series analysis, additional columns, which will not be used in the analysis, column types requiring conversion (e.g., character instead of factor), abbreviated variable names (e.g., BORO instead of BOROUGH), and minor value inconsistencies (e.g., "1022" must be recoded `NA` for the VIC_AGE_GROUP), the raw dataset required several preprocessing steps: employing `tidyverse`'s `select()`, `mutate()`, `rename()`, `group_by()`, and `summarize()` functions to obtain a modified tibble `NYPD`.

```
NYPD <- read_csv('./DATA/NYPD_Shooting_Incident_Data__Historic_.csv')        %>%
        select(c(OCCUR_DATE, BORO, PRECINCT, STATISTICAL_MURDER_FLAG,
                 VIC_AGE_GROUP, VIC_SEX, VIC_RACE))                          %>%
        mutate(  VIC_AGE_GROUP = na_if(VIC_AGE_GROUP, "1022"),
                 across(where(is.character), ~na_if(.x, "UNKNOWN")),
```

```
            across(where(is.character), ~na_if(.x, "(null)")),
            OCCUR_DATE              = mdy(OCCUR_DATE), BORO = factor(BORO),
            MONTH                   = floor_date(OCCUR_DATE, "month"),
            BORO                    = factor(str_to_title(BORO)),
            PRECINCT                = factor(PRECINCT),
            STATISTICAL_MURDER_FLAG = factor(STATISTICAL_MURDER_FLAG),
            VIC_AGE_GROUP           = factor(VIC_AGE_GROUP),
            VIC_SEX                 = factor(VIC_SEX),
            VIC_RACE                = factor(str_to_title(VIC_RACE)),
            SHOOTINGS               = 1)                                    %>%
  group_by(MONTH, BORO, PRECINCT, STATISTICAL_MURDER_FLAG,
                                  VIC_AGE_GROUP, VIC_SEX, VIC_RACE)         %>%
            summarize(SHOOTINGS     = sum(SHOOTINGS,  na.rm = TRUE))        %>%
  ungroup()                                                                 %>%
  rename(  BOROUGH                  = BORO)                                 %>%
  rename(  MURDER                   = STATISTICAL_MURDER_FLAG)              %>%
  rename(  VICTIM_AGE_GROUP         = VIC_AGE_GROUP)                        %>%
  rename(  VICTIM_SEX               = VIC_SEX)                              %>%
  rename(  VICTIM_RACE              = VIC_RACE)
```

The obtained tibble consists of 22,398 rows and 8 columns, where `MONTH` and `SHOOTINGS` were additionally created, with 123 missing values (i.e., a 99.9% reduction in the number of missing cells), corrected value labels, and readable variable names. To summarize the tibble `NYPD`, between January 2006 and December 2024, **29,744 cases** were reported, out of which 11,685 in Brooklyn, 8,834 in the Bronx, 3,977 in Manhattan, 4,426 in Queens, and 822 in Staten Island. **5,765** were identified as murders (**19.4%**), with the most victims in the **25-44 age group**, **male**, and **African Americans** in terms of race. The `summary()` output is below.

```
summary(NYPD)
```

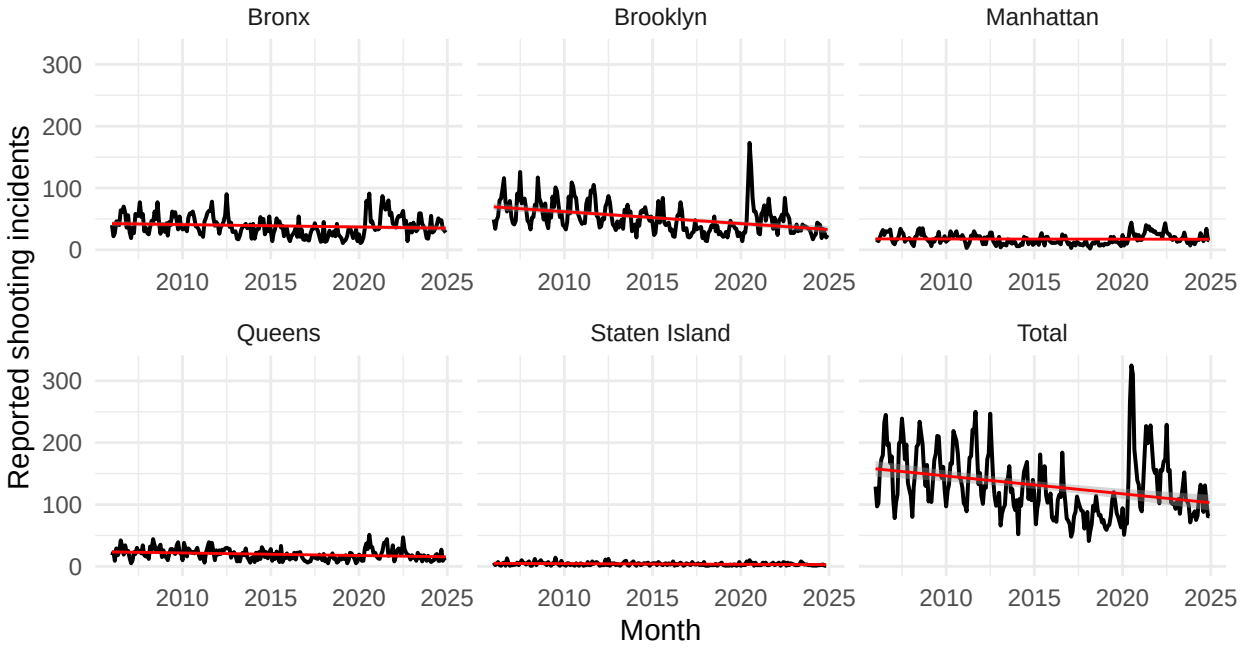
```
##      MONTH                       BOROUGH        PRECINCT       MURDER
##  Min.   :2006-01-01   Bronx        :6719   75     : 1033   FALSE:17551
##  1st Qu.:2010-01-01   Brooklyn     :8348   73     :  940   TRUE : 4847
##  Median :2014-07-01   Manhattan    :3208   44     :  868
##  Mean   :2014-12-16   Queens       :3483   67     :  814
##  3rd Qu.:2020-07-01   Staten Island: 640   46     :  762
##  Max.   :2024-12-01                        40     :  757
##                                            (Other):17224
##  VICTIM_AGE_GROUP VICTIM_SEX                          VICTIM_RACE
##  <18  :2538       F: 2574    American Indian/Alaskan Native:   12
##  18-24:7778       M:19813    Asian / Pacific Islander      :  413
##  25-44:9921       U:   11    Black                         :15005
##  45-64:1892                  Black Hispanic                : 2488
##  65+  : 209                  White                         :  675
##  NA's :  60                  White Hispanic                : 3742
##                              NA's                          :   63
##    SHOOTINGS
##  Min.   :1.000
##  1st Qu.:1.000
##  Median :1.000
##  Mean   :1.328
##  3rd Qu.:1.000
##  Max.   :9.000
##
```

Analysis of Time Trends

Aggregated monthly data show an uneven distribution of shooting incidents over time and across boroughs with a declining linear trend, where the steepest decline was observed in Brooklyn (see Table 1). Considering the spatial distribution of New York City's population, the Bronx (No. 4 by population) accounted for the second largest number of reported incidents (i.e., a higher rate per 1000 inhabitants), while Manhattan and Queens (No. 2 and 3) registered substantially fewer cases. Figure 1 also reveals periods of high and low shooting activity, including a citywide spike during the COVID-19 pandemic. This substantial increase may have been associated with socio-economic factors, such as social isolation during the government-imposed lockdowns, declining incomes, job losses, etc. – i.e., potential relationships that require further investigation.

```
ggplot(NYPD                                                                %>%
       group_by( MONTH, BOROUGH)                                           %>%
                summarize(SHOOTINGS  = sum(SHOOTINGS, na.rm = TRUE))       %>%
       ungroup()                                                           %>%
       bind_rows(NYPD                                                      %>%
                 group_by(MONTH)                                           %>%
                summarize(SHOOTINGS  = sum(SHOOTINGS, na.rm = TRUE),
                .groups          = "drop")                                %>%
                 mutate(BOROUGH  = "Total")
       ),       aes(x = MONTH, y = SHOOTINGS, group     = BOROUGH))          +
  geom_line(linewidth = 0.7) +
  geom_smooth(method = "lm", formula = y ~ x, linewidth = 0.5, color = "red")  +
  facet_wrap(~BOROUGH, axes = "all_x", axis.labels = "all_x")                  +
  scale_y_continuous(labels = scales::comma)                                   +
  labs(
    title = paste0("Figure 1. Reported shooting incidents, ",
                   as.character(m_min), "-", as.character(m_max)),
    x = "Month",
    y = "Reported shooting incidents"
  )                                                                            +
  theme_minimal()
```

Figure 1. Reported shooting incidents, January 2006−December 2024



The formal statistical test of **H1** is tied to coefficients β_1 from six linear regression models `lm(SHOOTINGS ~ MONTH SEQUENCE)`, i.e., for each borough and the dataset as a whole, with the `MONTH` encoded with the help of `seq_along()` into a time-series sequence. Table 1 comprises the slopes, their standard errors, the two-sided p-values, and the R^2 (coefficient of determination) of the fit. Apart from Manhattan, the remaining coefficients are non-zero at the 5% significance level ($\alpha = 0.05$), although the models reveal weak explanatory power – i.e., potential additional factors influencing the number of shooting incidents, such as a unit root.

```
NYPD                                                                    %>%
group_by( MONTH, BOROUGH)                                               %>%
         summarize(SHOOTINGS = sum(SHOOTINGS), .groups = "drop")        %>%
bind_rows(NYPD                                                          %>%
          group_by(MONTH) %>%
         summarize(SHOOTINGS = sum(SHOOTINGS), .groups = "drop")        %>%
          mutate(BOROUGH                              = "Total")
)                                                                       %>%
  filter(!is.na(BOROUGH),   !is.na(MONTH))                              %>%
 arrange(BOROUGH, MONTH) %>%
group_by(BOROUGH) %>%
         summarize(MODEL = list(summary(lm(SHOOTINGS ~ seq_along(MONTH))))) %>%
transmute(
         Borough = BOROUGH,
         Slope   = sapply(MODEL, \(m) m$coefficients[2, 1]),
         SE      = sapply(MODEL, \(m) m$coefficients[2, 2]),
         P       = sapply(MODEL, \(m) m$coefficients[2, 4]),
         R2      = sapply(MODEL, \(m) m$r.squared)
)                                                                       %>%
knitr::kable(digits  = 3,
             caption = "Linear trends in reported shooting incidents")
```

Table 1: Linear trends in reported shooting incidents

Borough	Slope	SE	P	R2
Bronx	-0.032	0.016	0.049	0.017
Brooklyn	-0.161	0.022	0.000	0.184
Manhattan	-0.002	0.008	0.788	0.000
Queens	-0.035	0.009	0.000	0.065
Staten Island	-0.009	0.003	0.001	0.051
Total	-0.241	0.049	0.000	0.097

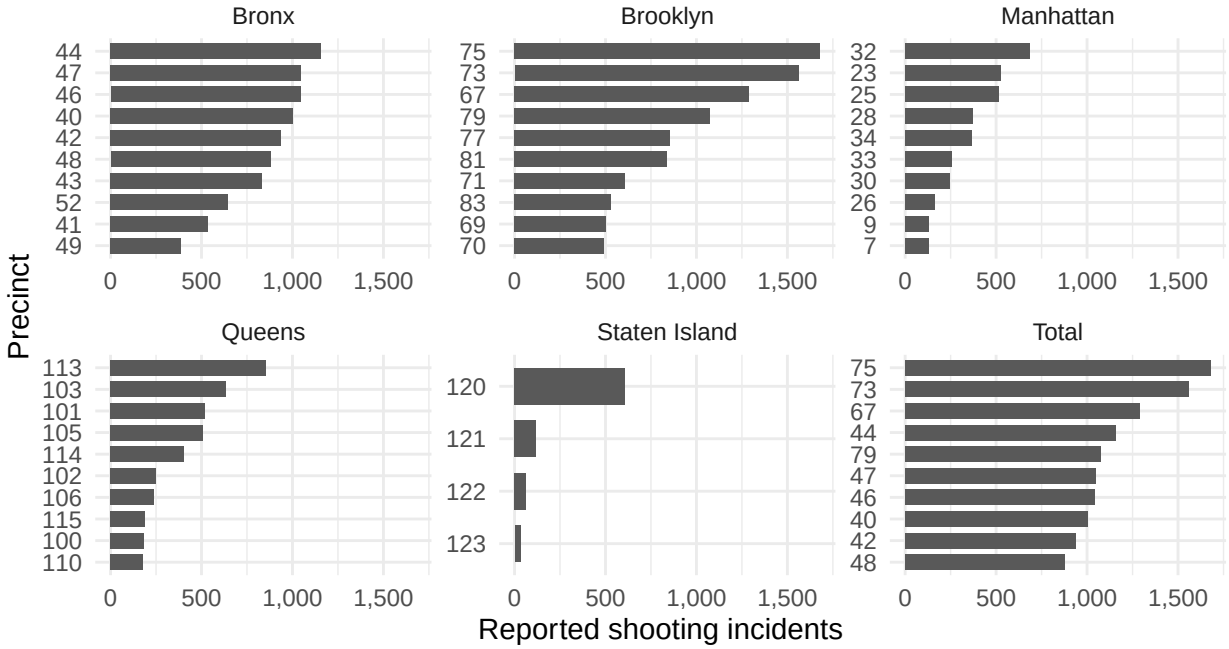
Analysis of Spatial Structure

Figure 2 demonstrates the distribution of shooting incidents by precinct. Namely, the three precincts with the most reported incidents are **No. 75, 73, 67** citywide, **No. 44, 47, 46** in the Bronx, (the above-mentioned) **No. 75, 73, 67** in Brooklyn, **No. 32, 23, 25** in Manhattan, **No. 113, 103, 101** in Queens, and **No. 120, 121, 122** in Staten Island. These findings strongly indicate a non-uniform distribution of the examined data.

```
ggplot(NYPD                                                             %>%
       group_by( BOROUGH, PRECINCT)                                     %>%
                summarize(SHOOTINGS  = sum(SHOOTINGS,       na.rm = TRUE)) %>%
       ungroup()                                                        %>%
       bind_rows(NYPD                                                   %>%
                 group_by(PRECINCT)                                     %>%
                summarize(SHOOTINGS  = sum(SHOOTINGS,       na.rm = TRUE),
                .groups         = "drop")                               %>%
```

```
                mutate(BOROUGH = "Total")
      )                                                                    %>%
      group_by( BOROUGH)                                                   %>%
               slice_max(order_by = SHOOTINGS, n = 10, with_ties = FALSE)  %>%
      ungroup(),    aes(x = reorder(PRECINCT, SHOOTINGS), y = SHOOTINGS))    +
  geom_col(width = 0.7)                                                    +
  coord_flip()                                                             +
  facet_wrap(~ BOROUGH, scales  = "free_y", axes        = "all_x",
                                            axis.labels = "all_x")         +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = paste0("Figure 2. Top ten precincts by incidents, ",
                   as.character(m_min), "-", as.character(m_max)),
    x = "Precinct",
    y = "Reported shooting incidents"
  )                                                                        +
  theme_minimal()
```

Figure 2. Top ten precincts by incidents, January 2006−December 2024



The formal statistical test of **H2** is based on Pearson's chi-square goodness-of-fit test `chisq.test(SHOOTINGS)` applied to six distributions of reported shooting incidents across precincts, i.e., for each borough and the dataset as a whole, with the null hypothesis assuming a uniform distribution of incidents ($H_0 : p_1 = p_2 = \cdots = p_k = 1/k$), where k is the number of precincts, and a p-value precision of 0.001. The resulting Table 2 comprises the χ^2 statistics, degrees of freedom, and upper-tail p-values. H_0 is rejected in all cases at the 5% significance level ($\alpha = 0.05$), confirming a statistically significant non-uniform distribution of shooting incidents across precincts, which may reflect variations in population density, socio-economic conditions, etc. – i.e., potential relationships meriting further investigation with the help of statistical (quantitative) modeling.

```
NYPD                                                                   %>%
group_by( BOROUGH, PRECINCT)                                           %>%
         summarize(SHOOTINGS = sum(SHOOTINGS), .groups = "drop")       %>%
bind_rows(NYPD                                                         %>%
```

```
          group_by(PRECINCT)                                                    %>%
         summarize(SHOOTINGS = sum(SHOOTINGS), .groups = "drop")                %>%
          mutate(BOROUGH                                = "Total")
)                                                                               %>%
  filter(!is.na(BOROUGH),   !is.na(PRECINCT))                                   %>%
group_by( BOROUGH)                                                              %>%
         summarize(TEST = list(chisq.test(SHOOTINGS)))                          %>%
transmute(
         Borough = BOROUGH,
         Chi2    = sapply(TEST, \(m) unname(m$statistic)),
         DF      = sapply(TEST, \(m) unname(m$parameter)),
         P       = sapply(TEST, \(m) m$p.value)
)                                                                               %>%
mutate(P = scales::pvalue(P, accuracy = 0.001))                                 %>%
knitr::kable(digits  = 3,
             caption = "Pearson chi-square tests of uniform distribution")
```

Table 2: Pearson chi-square tests of uniform distribution

Borough	Chi2	DF	P
Bronx	1755.455	11	<0.001
Brooklyn	10387.570	22	<0.001
Manhattan	4368.899	21	<0.001
Queens	3129.073	15	<0.001
Staten Island	1068.696	3	<0.001
Total	29788.618	76	<0.001

Incident Classification Model

In order to identify the probability and corresponding odds of a shooting incident being classified as a murder, – which may provide valuable insights for police precincts in allocating resources between calls, anticipating risks, or developing potential preventive measures – the following logistic regression (a generalized linear model of the binomial family with a logit link) is employed, with the log-odds modeled as a function of `PRECINCT` (`P`), `VICTIM_AGE_GROUP` (`A`), `VICTIM_SEX` (`S`), and `VICTIM_RACE` (`R`). Its predictors (features) were selected to reflect geography – given the above-established non-uniformity of distribution of shooting incidents across precincts – and demographics, specifically victim characteristics, given the limited availability of information (a higher proportion of missing values) on other potentially relevant predictors in the original (raw) dataset.

$$\log\left(\frac{P(\text{MURDER}=1)}{1-P(\text{MURDER}=1)}\right) = \beta_0 + \sum_{j=1}^{J_P}\beta_{P_j}P_j + \sum_{j=1}^{J_A}\beta_{A_j}A_j + \sum_{j=1}^{J_S}\beta_{S_j}S_j + \sum_{j=1}^{J_R}\beta_{R_j}R_j$$

The results of the `glm()` function estimated on the tibble `NYPD`, aggregated across `BOROUGH` and `MONTH` with `SHOOTINGS` set as frequency weights, are presented in Tables 3 and 4, and the corresponding R code below.

```
model <- glm(
  MURDER ~         PRECINCT + VICTIM_AGE_GROUP + VICTIM_SEX + VICTIM_RACE,
  data = NYPD                                                                   %>%
         group_by(PRECINCT,  MURDER,
                             VICTIM_AGE_GROUP,  VICTIM_SEX,  VICTIM_RACE)     %>%
                  summarise(SHOOTINGS = sum(SHOOTINGS), .groups = "drop"),
  weights = SHOOTINGS,
```

```
  family = binomial()
)
```

Following the model, the formal statistical test of **H3** is $H_0 : \forall_{\beta_{x_j}} \ \beta_{x_j} = 0$ against $H_1 : \exists_{\beta_{x_j}} \ \beta_{x_j} \neq 0$, where $x_j \in \{P_j, A_j, S_j, R_j\}$. Table 3 comprises the statistically significant coefficients β_{x_j} (log-odds) and equivalent odds ratios $exp(\beta_{x_j})$ at the 5% significance level ($\alpha = 0.05$) (i.e., the table is truncated given the number of precincts in the dataset), their standard errors, and the two-sided p-values. Individual β_{x_j} indicate significant odds differences across police precincts and victim age groups. On the one hand, relative to Precinct 1 (the reference category), 28 precincts have significantly lower odds of a murder incident, with odds ratios ranging from 0.164 (Precinct 7) to 0.448 (Precinct 73) and standard errors being near-identical ($\approx 0.40 - 0.60$). On the other hand, victim age group shows a clear increasing pattern: relative to victims under 18 (the reference category), the odds of a murder incident are ca. 29%, 81%, 111%, and 169% higher for ages 18–24, 25–44, 45–64, and 65+ respectively, with significantly lower standard errors ($\approx 0.06 - 0.15$). No level of victim sex or race, included in the logistic regression equation, was found to be statistically significant at the 5% level.

```
as.data.frame(summary(model)$coefficients)                                %>%
tibble::rownames_to_column("Predictor")                                   %>%
filter(`Pr(>|z|)` < 0.05)                                                 %>%
transmute(
  Predictor,
  `Coefficient (log-odds)` = Estimate,
  `Odds ratio`             = exp(Estimate),
  `Std. error`             = `Std. Error`,
  `P`                      = scales::pvalue(`Pr(>|z|)`, accuracy = 0.001)
)                                                                         %>%
knitr::kable(
  digits  = 3,
  caption = "Statistically significant logistic regression coefficients"
)
```

Table 3: Statistically significant logistic regression coefficients

Predictor	Coefficient (log-odds)	Odds ratio	Std. error	P
PRECINCT7	-1.806	0.164	0.514	<0.001
PRECINCT20	-1.452	0.234	0.589	0.014
PRECINCT23	-0.964	0.381	0.411	0.019
PRECINCT25	-0.889	0.411	0.411	0.030
PRECINCT26	-1.377	0.252	0.464	0.003
PRECINCT28	-0.950	0.387	0.418	0.023
PRECINCT32	-0.826	0.438	0.406	0.042
PRECINCT34	-1.026	0.358	0.419	0.014
PRECINCT40	-0.934	0.393	0.403	0.020
PRECINCT43	-0.869	0.419	0.404	0.032
PRECINCT66	-1.156	0.315	0.551	0.036
PRECINCT68	-1.176	0.309	0.600	0.050
PRECINCT71	-0.997	0.369	0.409	0.015
PRECINCT73	-0.803	0.448	0.399	0.044
PRECINCT76	-0.910	0.403	0.438	0.038
PRECINCT78	-1.350	0.259	0.532	0.011
PRECINCT79	-0.814	0.443	0.402	0.043
PRECINCT81	-0.828	0.437	0.404	0.040
PRECINCT83	-0.876	0.416	0.410	0.032
PRECINCT84	-1.063	0.345	0.460	0.021
PRECINCT90	-0.894	0.409	0.418	0.033

Predictor	Coefficient (log-odds)	Odds ratio	Std. error	P
PRECINCT94	-1.303	0.272	0.502	0.009
PRECINCT100	-1.614	0.199	0.469	<0.001
PRECINCT101	-0.871	0.419	0.411	0.034
PRECINCT103	-1.033	0.356	0.409	0.011
PRECINCT108	-1.530	0.217	0.530	0.004
PRECINCT114	-0.971	0.379	0.415	0.019
PRECINCT121	-0.929	0.395	0.462	0.044
VICTIM_AGE_GROUP18-24	0.253	1.288	0.060	<0.001
VICTIM_AGE_GROUP25-44	0.594	1.812	0.058	<0.001
VICTIM_AGE_GROUP45-64	0.745	2.107	0.074	<0.001
VICTIM_AGE_GROUP65+	0.988	2.685	0.153	<0.001

Furthermore, the deviance-based likelihood-ratio (LR) test rejects the null hypothesis of joint insignificance of the included predictors $\chi^2(87) = 435.603$, with a p-value smaller than 0.001, providing further statistical support for `H_1` (see Table 4). Still, McFadden's pseudo-R^2 of 0.015 indicates only a ca. 1.5% improvement in log-likelihood relative to the intercept-only model, suggesting limited explanatory power and reducing the practical significance of the findings. Thus, additional features should be considered in further investigation.

```
tibble(
  Statistic = c(
    "Observations", "Null deviance", "Residual deviance", "LR chi-square",
    "LR degrees of freedom", "LR p-value", "AIC", "McFadden pseudo-R²"
  ),
  Value = c(
    sum(weights(model, type = "prior")),
    round(model$null.deviance, 3), round(model$deviance, 3),
    round(model$null.deviance - model$deviance,          3),
    model$df.null -  model$df.residual, scales::pvalue(
      pchisq(model$null.deviance - model$deviance,
             df = model$df.null  - model$df.residual,    lower.tail = FALSE),
      accuracy  = 0.001
    ),                                 round(AIC(model),     3),
    round(1 -     model$deviance / model$null.deviance,  3)
  )
)                                                                    %>%
knitr::kable(
  digits  = 3,
  caption = "Logistic regression goodness-of-fit and likelihood-ratio test"
)
```

Table 4: Logistic regression goodness-of-fit and likelihood-ratio test

Statistic	Value
Observations	29624
Null deviance	29139.609
Residual deviance	28704.006
LR chi-square	435.603
LR degrees of freedom	87

Statistic	Value
LR p-value	<0.001
AIC	28880.006
McFadden pseudo-R^2	0.015

Discussion of Potential Bias

The findings in the report, despite the use of rigorous methodology, are subject to several potential biases, the first group of which is data-related. On the one hand, the original (raw) dataset comprises 94,426 missing cells, to which the recoding of placeholders (e.g., "UNKNOWN", "(null)", and "1022") added further `NA` values, namely, in the victim characteristics. Since statistical models, including `glm()`, dropped rows with missing values by default, the logistic regression was estimated on a smaller subset of victims (29,624, see Table 4), which might have introduced selection bias if the missing values were not randomly distributed, in addition to the loss of information for estimation. On the other hand, aggregation to monthly frequency discarded the within-month and hourly variability (`OCCUR_TIME`), while aggregation across `BOROUGH` and `MONTH` for the classification model assumed constant odds between January 2006 and December 2024. Moreover, the data itself might also have been a source of bias due to how it was collected and compiled into the original dataset.

The second group of potential biases is method-related, given that the three statistical hypotheses were tested with basic (textbook) procedures, and more advanced models could be employed. First, the linear trends in Table 1 did not account for potential unit roots, seasonality of `SHOOTINGS`, and violations of Gauss-Markov assumptions, which might have produced biased or spurious estimates – e.g., a (seasonal) ARIMA could be a more adequate model. Second, given the data limitations, the logistic regression included only four predictors (and levels thereof) and its McFadden's pseudo-R^2 of 0.015 hinted at possible omission of relevant predictors, potentially contributing to the bias. Hence, although these limitations do not necessarily invalidate the findings, they merit further statistical investigation (which is, however, beyond the scope of this short report).

Conclusion

This brief report examined 29,744 shooting incidents recorded by the NYPD, which took place between January 2006 and December 2024 and out of which ca. 19.4% were classified as murders, and answered the three research questions with the help of formal statistical hypotheses tested at the 5% significance level ($\alpha = 0.05$). First, in terms of long-term trends (1), monthly time series exhibited a declining linear trend for the City of New York as a whole and for four of the five boroughs, with the steepest slope (i.e., decline) in Brooklyn, leading to acceptance of **H1** in all cases except Manhattan (see Table 1) – although the low R^2 values and a citywide spike in incidents during the COVID-19 pandemic (see Figure 1) merit further investigation. Second, in terms of the spatial structure (2), shooting incidents were found to be concentrated in a small number of precincts, namely, in the Bronx and Brooklyn (see Figure 2), and Pearson's χ^2 goodness-of-fit tests rejected the uniform distribution for each borough and the dataset as a whole (see Table 2), i.e., leading to rejection of **H2**. Third, in terms of murder classification (3), the logistic regression statistically supported **H3**: the odds of a murder incident differed significantly across precincts (e.g., 28 precincts exhibited lower odds when compared to Precinct 1) and increased with victim age (up to 169% higher odds when compared to victims under 18), while victim sex and race were not found to be statistically significant (see Table 3), a finding also confirmed by an LR test (see Table 4). Still, McFadden's pseudo-R^2 of 0.015 suggested limited explanatory power of geography and victim demographics. To sum up, in light of the potential data-related and method-related biases, the findings of the report should be interpreted with caution; hence, the use of more advanced time-series models, as well as additional predictors, is recommended.

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I acknowledge the use of Grammarly (https://www.grammarly.com) to check the spelling, grammar, and punctuation of the final text of this report.